

CODE:-

expense tracker.py

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```
1  expenses = []
2
3  def add_expense():
4      name = input("Enter expense name: ")
5      category = input("Enter category (Food/Travel/Shopping/etc): ")
6      amount = float(input("Enter amount: "))
7
8      expense = {
9          "name": name,
10         "category": category,
11         "amount": amount
12     }
13
14     expenses.append(expense)
15     print("Expense added successfully!")
16
17 def view_expenses():
18     if len(expenses) == 0:
19         print("No expenses found.")
20         return
21
22     print("\n----- ALL EXPENSES -----")
23
24     for i, expense in enumerate(expenses, start=1):
25         print(f"{i}. {expense['name']} | {expense['category']} | ₹{expense['amount']}")
26
27 def total_expense():
28     total = 0
29
30     for expense in expenses:
31         total += expense["amount"]
32
33     print(f"\nTotal Spending = ₹{total}")
34
35 def average_expense():
36     if len(expenses) == 0:
37         print("No expenses available.")
```

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34
35 def average_expense():
36     if len(expenses) == 0:
37         print("No expenses available.")
38         return
39
40     total = sum(expense["amount"] for expense in expenses)
41
42     avg = total / len(expenses)
43
44     print(f"Average Expense = ₹{avg:.2f}")
45
46 def highest_expense():
47     if len(expenses) == 0:
48         print("No expenses available.")
49         return
50
51     highest = max(expenses, key=lambda x: x["amount"])
52
53     print("\nHighest Expense:")
54     print(f"{highest['name']} | {highest['category']} | ₹{highest['amount']}")
55
56 def search_expense():
57     keyword = input("Enter expense name to search: ").lower()
58
59     found = False
60
61     for expense in expenses:
62         if keyword in expense["name"].lower():
63             print(f"{expense['name']} | {expense['category']} | ₹{expense['amount']}")
64             found = True
65
66     if not found:
67         print("Expense not found.")
68
69 def delete_expense():
70     view_expenses()
```

```
1 # Expense Tracker
2 """
3 This program tracks expenses and provides various functions to manage them.
4 It uses a list to store expense data and provides functions to view, add,
5 delete, search, and calculate average and highest expenses.
6 """
7
8 # Global variable to store expenses
9 expenses = []
10
11 # Function to view expenses
12 def view_expenses():
13     """View the list of expenses"""
14     if len(expenses) == 0:
15         print("No expenses available.")
16     else:
17         for expense in expenses:
18             print(f"{expense['name']} | {expense['category']} | ₹{expense['amount']}")
19
20 # Function to add expense
21 def add_expense():
22     """Add a new expense"""
23     name = input("Enter expense name: ")
24     category = input("Enter expense category: ")
25     amount = float(input("Enter expense amount: "))
26     expense = {"name": name, "category": category, "amount": amount}
27     expenses.append(expense)
28     print("Expense added successfully.")
29
30 # Function to delete expense
31 def delete_expense():
32     """Delete an expense"""
33     view_expenses()
34     if len(expenses) == 0:
35         print("No expenses available.")
36         return
37     index = int(input("Enter the index of the expense to delete: "))
38     if 0 <= index < len(expenses):
39         del expenses[index]
40         print("Expense deleted successfully.")
41     else:
42         print("Invalid index.")
43
44 # Function to search expense
45 def search_expense():
46     """Search for an expense by name"""
47     keyword = input("Enter expense name to search: ").lower()
48     found = False
49     for expense in expenses:
50         if keyword in expense["name"].lower():
51             print(f"{expense['name']} | {expense['category']} | ₹{expense['amount']}")
52             found = True
53     if not found:
54         print("Expense not found.")
55
56 # Function to calculate average expense
57 def average_expense():
58     """Calculate the average expense"""
59     if len(expenses) == 0:
60         print("No expenses available.")
61         return
62     total = sum(expense["amount"] for expense in expenses)
63     avg = total / len(expenses)
64     print(f"Average Expense = ₹{avg:.2f}")
65
66 # Function to find highest expense
67 def highest_expense():
68     """Find the highest expense"""
69     if len(expenses) == 0:
70         print("No expenses available.")
71         return
72     highest = max(expenses, key=lambda x: x["amount"])
73     print("\nHighest Expense:")
74     print(f"{highest['name']} | {highest['category']} | ₹{highest['amount']}")
75
76 # Main program
77 if __name__ == "__main__":
78     while True:
79         print("\nExpense Tracker")
80         print("1. View Expenses")
81         print("2. Add Expense")
82         print("3. Delete Expense")
83         print("4. Search Expense")
84         print("5. Average Expense")
85         print("6. Highest Expense")
86         print("7. Exit")
87         choice = input("Enter your choice: ")
88         if choice == "1":
89             view_expenses()
90         elif choice == "2":
91             add_expense()
92         elif choice == "3":
93             delete_expense()
94         elif choice == "4":
95             search_expense()
96         elif choice == "5":
97             average_expense()
98         elif choice == "6":
99             highest_expense()
100        elif choice == "7":
101            break
102
```

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```
68
69 def delete_expense():
70     view_expenses()
71
72     try:
73         index = int(input("Enter expense number to delete: ")) - 1
74
75         removed = expenses.pop(index)
76
77         print(f"{removed['name']} deleted successfully!")
78
79     except:
80         print("Invalid expense number.")
81
82 while True:
83
84     print("\n===== EXPENSE TRACKER =====")
85     print("1. Add Expense")
86     print("2. View Expenses")
87     print("3. Total Spending")
88     print("4. Average Expense")
89     print("5. Highest Expense")
90     print("6. Search Expense")
91     print("7. Delete Expense")
92     print("8. Exit")
93
94     choice = input("Enter your choice: ")
95
96     if choice == "1":
97         add_expense()
98
99     elif choice == "2":
100         view_expenses()
101
102     elif choice == "3":
103         total_expense()
104
```

```
68
69 def delete_expense():
70     view_expenses()
71
72     try:
73         index = int(input("Enter expense number to delete: ")) - 1
74
75         removed = expenses.pop(index)
76
77         print(f"{removed['name']} deleted successfully!")
78
79     except:
80         print("Invalid expense number.")
81
82 while True:
83
84     print("\n===== EXPENSE TRACKER =====")
85     print("1. Add Expense")
86     print("2. View Expenses")
87     print("3. Total Spending")
88     print("4. Average Expense")
89     print("5. Highest Expense")
90     print("6. Search Expense")
91     print("7. Delete Expense")
92     print("8. Exit")
93
94     choice = input("Enter your choice: ")
95
96     if choice == "1":
97         add_expense()
98
99     elif choice == "2":
100         view_expenses()
101
102     elif choice == "3":
103         total_expense()
104
```

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90     print("6. Search Expense")
91     print("7. Delete Expense")
92     print("8. Exit")
93
94     choice = input("Enter your choice: ")
95
96     if choice == "1":
97         add_expense()
98
99     elif choice == "2":
100         view_expenses()
101
102     elif choice == "3":
103         total_expense()
104
105     elif choice == "4":
106         average_expense()
107
108     elif choice == "5":
109         highest_expense()
110
111     elif choice == "6":
112         search_expense()
113
114     elif choice == "7":
115         delete_expense()
116
117     elif choice == "8":
118         print("Thank you for using Expense Tracker!")
119         break
120
121     else:
122         print("Invalid choice. Please try again.")
```

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OUTPUT:-

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [ ] ... | [ ] [ ] X

PS C:\Users\User\OneDrive\Desktop\Desktop\AnanyaProject> & C:/Users/User/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/User/OneDrive/Desktop/Desktop/AnanyaProject/expense_tracker.py"

===== EXPENSE TRACKER =====
1. Add Expense
2. View Expenses
3. Total Spending
4. Average Expense
5. Highest Expense
6. Search Expense
7. Delete Expense
8. Exit
Enter your choice:
```

PROJECT REPORT

Project Title:

Expense Tracker

Intern Name:

Ananya Jain

Organization:

DecodeLabs

Technology Used:

Python

Project Objective:

The objective of this project is to develop an Expense Tracker application that allows users to record, manage, and analyze their daily expenses.

Features:

- Add expenses
- View expense history
- Search expenses
- Delete expenses
- Calculate total spending
- Calculate average spending
- Display highest expense
- Save records to file

Python Concepts Used:

- Lists
- Dictionaries
- Functions
- Loops
- Conditional Statements
- File Handling

Project Working:

Users can enter expense details along with categories. The application stores expenses in a list and performs calculations such as total, average, and highest expense. Users can also search and manage records efficiently.

Learning Outcomes:

- Understanding of data accumulation
- Practical use of mathematical operations
- Experience with file handling
- Improved problem-solving skills

Conclusion:

The Expense Tracker demonstrates the practical application of Python programming concepts for financial data management and analysis.